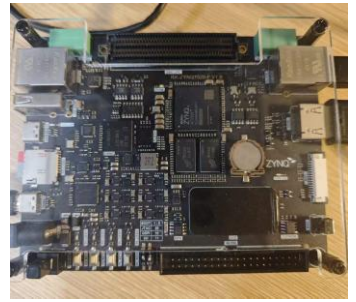
Copyright 2026

# One Frame, Two Truths
# A Multi-Source Video Node on Zynq-7020

**Wang Xu · Uie_Team**

Advisors: Tan Huisheng · Liang Jiaojiao
Hunan University of Technology, Hunan Province
Platform: RK-ZYNQ7020-F / xc7z020clg484-2



*On-board test · system output*

## INTRODUCTION

Remote video nodes usually have one Ethernet cable and no PC. The operator needs to know whether the frame arrived, whether processing took effect, and how old the picture is. This node performs the path on one Zynq-7020: a 512×300 stream enters through GbE, whole frames are committed after three conditions, and both panes read back the same frame.

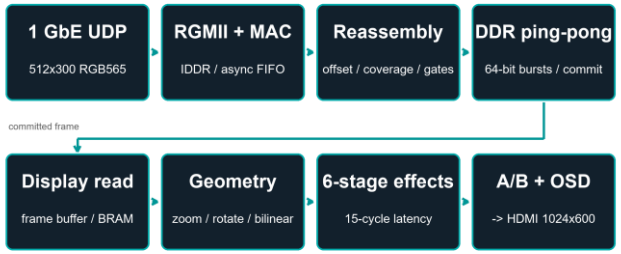


### Typical application case

Confidence monitoring for broadcast, security and industrial vision: one cable carries the picture, processing result and link status.

### Key technical issues addressed

1. GbE UDP reassembly: offset-based reassembly; commit only when bytes, row coverage and bad-packet count pass.
2. Same-frame comparison: both panes share geometry and a fixed 15-cycle effect chain; the seam selects per pixel.
3. Status monitoring: aborts, missing rows, frame gaps and latency are readable through serial, telemetry and GPIO.
4. Storage: DDR ping-pong frame buffer + on-board test card.




## SYSTEM ARCHITECTURE

### Ingress & storage

1. RGMII → MAC → UDP filter → offset reassembly;
2. Commit only when bytes, row coverage and bad-packet count pass;
3. PL writes DDR ping-pong as AXI HP0 master.

### Display & processing

1. Read back the frame; geometry, bilinear and six-stage effects;
2. Same-frame comparison: left original, right processed, per-pixel seam;
3. OSD + HDMI 1024×600; status via serial/telemetry/GPIO.

## CREATIVE DESING

## RESULT

### CREATIVE DESIGN

1. Same-frame comparison: both panes share geometry, row labels and the fixed 15-cycle chain; the seam compares one frame.
2. Commit conditions + status monitoring: bytes, row coverage and bad-packet count must pass; aborts, missing rows and frame gaps explain failures.

### RESULT

RGMII → HDMI entirely in PL; PS only control. 1024×600 @ 59.5 Hz.
Timing: WNS 1.368 / WHS 0.048 ns, zero failures.
8 s stream: 241 frames @ 30.01 fps; selftest 22.2/0/18.9 PASS.
UART 105 commands PASS; QSPI boot validated.
Next: DDR tile cache / 1080p link / RGMII eye diagram / remote update.